

F - Critical Misread

分值: 525 分

Problem Statement

You are given K strings S_i consisting of lowercase English letters.

Find the number, modulo 998244353, of strings of length N consisting of lowercase English letters that do not contain any of $S_1, S_2, \dots, S_K$ as a substring (contiguous subsequence).

给定 K 个由小写英文字母组成的字符串 S_i 。求所有长度为 N 、由小写英文字母组成，且不以任何 $S_1, S_2, \dots, S_K$ 作为子串（连续子序列）的字符串的数量，答案对 998244353 取模。

Constraints

- N is an integer between 1 and 10^9 , inclusive.
 - K is an integer between 1 and 10, inclusive.
 - S_i is a string consisting of lowercase English letters with length between 1 and 10, inclusive.
 - N 是一个介于 1 和 10^9 之间的整数（包含两端）。
 - K 是一个介于 1 和 10 之间的整数（包含两端）。
 - S_i 是一个由小写英文字母组成的字符串，长度在 1 到 10 之间（包含两端）。
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Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N K

S_1

S_2

$\vdots$

S_K

Output

Output the answer.

输出答案。

Sample Input 1

```
3 2
aa
ab
```

Sample Output 1

```
17474
```

- The number of strings of length 3 consisting of lowercase English letters is $26^3 = 17576$.
- The number of strings of length 3 consisting of lowercase English letters whose first two characters are aa is 26.
- The number of strings of length 3 consisting of lowercase English letters whose last two characters are aa is 26.
- The number of strings of length 3 consisting of lowercase English letters whose first two characters are ab is 26.
- The number of strings of length 3 consisting of lowercase English letters whose last two characters are ab is 26.
- The number of strings of length 3 consisting of lowercase English letters whose entirety is aaa is 1.
- The number of strings of length 3 consisting of lowercase English letters whose entirety is aab is 1.
- Combining the above facts, it can be confirmed that the answer for this test case is 17474.
- 长度为 3、由小写英文字母组成的字符串共有 $26^3 = 17576$ 个。
- 长度为 3、由小写英文字母组成，且前两个字符为 aa 的字符串共有 26 个。
- 长度为 3、由小写英文字母组成，且最后两个字符为 aa 的字符串共有 26 个。

- 长度为 3、由小写英文字母组成，且前两个字符为 ab 的字符串共有 26 个。
- 长度为 3、由小写英文字母组成，且最后两个字符为 ab 的字符串共有 26 个。
- 长度为 3、由小写英文字母组成，且整个串为 aaa 的字符串共有 1 个。
- 长度为 3、由小写英文字母组成，且整个串为 aab 的字符串共有 1 个。
- 结合以上信息，可以确认该测试用例的答案为 17474。

Sample Input 2

```
1 1
ab
```

Sample Output 2

```
26
```

Sample Input 3

```
1000000000 10
contest
tester
error
orange
angel
elegant
antitese
sextuple
pleasure
surely
```

Sample Output 3

```
698570468
```

Find the count modulo 998244353.

求数量对 998244353 取模的结果。